Machine Learning Based Analog Circuit Fault Diagnosis

PDF 12 — Chapter 19: Dashboard Architecture

Complete Software Architecture | Every Button Explained

PART 1 — Complete Software Architecture

Chapter 19 — Dashboard Architecture Part 1 — Complete Software Architecture (From User Upload to Fault Prediction) Before Starting Congratulations. Until now, our project looked like this. **LTspice** — CSV Dataset.

Feature Extraction

SVM / ANN — Prediction. Question. Can a normal user use this? No. A normal user cannot open Python, run scripts, load models, and extract features. Need an interface. That interface is called the Dashboard. What Is a Dashboard?

A Dashboard

is simply a Graphical User Interface (GUI) that allows a user to interact with our Machine Learning model without writing code.

PROFESSOR ASKS

What is the purpose of the dashboard?

EXCELLENT ANSWER

The dashboard provides a user-friendly interface that allows users to upload waveform data, automatically extract features, perform fault classification using the trained model, and visualize the results without interacting directly with the Python code. Excellent. Why Build a Dashboard? Question. Suppose our project contains 2000 lines of Python code. Question. Should the user read all those lines? No. Instead they should see a simple interface. Example

Upload File

Click Predict

View Results

Very simple.

Real Life Analogy

Imagine Google. Question. Do you see its millions of lines of code? No. You only see

Search Box

Exactly the same idea. The dashboard hides the complexity.

Dashboard Philosophy

Question. Does the dashboard perform Machine Learning? No. Very important. The dashboard only controls the workflow. The real work is still done by Feature Extraction StandardScaler SVM ANN The dashboard simply connects everything together.

Complete Architecture

Our dashboard follows this pipeline. **User** — Upload CSV.

Feature Extraction

StandardScaler

Load Saved Model

Prediction

Display Result

Notice the dashboard is like a manager. It tells each module when to work.

Step 1 — User Upload

Everything starts here. The user clicks Upload and selects a waveform or CSV file. Question. Has

Machine Learning

started? No. The dashboard has only received the file.

PROFESSOR ASKS

What happens immediately after the user uploads a file?

EXCELLENT ANSWER

The uploaded file is received by the application, validated, and prepared for feature extraction. No machine learning prediction occurs at this stage.

Step 2 — Input Validation

Question. Suppose the user uploads movie.mp4 Question. Can our system classify it? Obviously not. Need validation. The dashboard checks Correct file type Correct format Correct columns Missing values Empty files Only then does it continue.

Why Validation Matters

Without validation, our program could crash before reaching the classifier. Good software always checks the input first.

Step 3 — Feature Extraction

Now the dashboard calls our Feature Extraction module. Question. Does the dashboard calculate Peak, RMS, Energy, DC Level? No. It simply calls the function that already exists. That function returns our 17-dimensional feature vector. Pipeline becomes CSV

Feature Extraction

17 Features Step 4 StandardScaler Question. Can we directly send the features to SVM? No. Remember Chapter 16. Need StandardScaler. Question. Does the dashboard fit a new scaler? No. Very important. It loads the already-trained scaler. Exactly the same scaler used during training.

PROFESSOR ASKS

Why must the same StandardScaler be used during prediction?

EXCELLENT ANSWER

The same StandardScaler used during training must also be used during prediction so that the input features are transformed using the identical scaling parameters. This ensures consistency between training and inference.

Step 5 — Loading the Model

Question. Does the dashboard train the SVM every time? No. That would be extremely slow. Instead the dashboard loads the already-trained model. Conceptually **Saved SVM** — Load. Ready Training never happens inside the dashboard.

Step 6 — Prediction

Now everything is ready. The dashboard sends the scaled feature vector to the classifier. Example

17 Features

SVM — Prediction. RC_Open Simple.

PROFESSOR ASKS

What input is given to the trained SVM during prediction?

EXCELLENT ANSWER

The trained SVM receives the scaled 17-dimensional feature vector extracted from the uploaded waveform and predicts the corresponding fault class.

Step 7 — Displaying the Result

Now the dashboard shows the prediction. Example

Predicted Fault

RC_Open Optionally it may also display Confidence score Waveform Extracted Features Processing Time depending on the implementation.

User Experience

Notice how simple the process looks to the user. **Upload** — Predict. Result Behind the scenes, however, many operations are happening. Behind the Scenes **Upload** — Validation.

Feature Extraction

Scaling

Load Model

Prediction — Display. This is what actually happens.

Why This Architecture Is Good

Every module has one responsibility.

Feature Extraction

Extract Features. **Scaler** — Normalize Features. **SVM** — Classify. **Dashboard** — Coordinate Everything. This makes the software easy to maintain and improve.

Real Engineering Principle

Notice our project uses modular design. Question. Suppose tomorrow we replace SVM with Random Forest. Question. Do we need to rewrite the dashboard? No. Only the classifier module changes. Everything else remains the same. This is good software engineering.

Connection to Our Project

Our complete software architecture now becomes **User** — Dashboard.

Input Validation

Feature Extraction

StandardScaler

Load Trained Model

Prediction

Display Results

This is exactly how our application works.

COMMON MISTAKES

MISTAKE: [X] The dashboard trains the model. Wrong.

MISTAKE: [X] The dashboard extracts features by itself. Wrong. It calls the feature extraction module.

MISTAKE: [X] Prediction is made before scaling. Wrong. Scaling must happen first.

MISTAKE: [X] The dashboard contains all the machine learning logic. Wrong. It only coordinates the modules.

REVISION NOTES

Remember
the dashboard workflow.
Upload — Receive File.
Validation — Check Input.
Feature Extraction — 17 Features.
Scaling — Normalize.
Model — Predict.
Dashboard — Display Result.

MEMORY TRICK

Remember this sentence. "Upload, Validate, Extract, Scale, Predict, Display." Six words. Entire dashboard pipeline.

KNOWLEDGE CHECK

Level 1

- What is the purpose of the dashboard?
- What happens after file upload?
- Why is input validation necessary?
- Why do we use the saved StandardScaler?
- Does the dashboard train the model?

Level 2

- Explain the complete dashboard workflow.
- Why is modular design beneficial?
- Why is feature extraction separated from the dashboard?
- Why is prediction performed only after scaling?
- How does the dashboard improve usability?

Professor Level

- Explain the complete software architecture of your fault diagnosis system.
- Why is separating the GUI from the machine learning logic considered good software engineering?
- Describe the data flow from user upload to final prediction.
- How does your dashboard ensure consistency between training and deployment?
- If you replaced SVM with another classifier, what parts of the dashboard would remain unchanged?
- End of Chapter 19 (Part 1)
- Congratulations.
- You now understand the complete architecture of your application.
- More importantly, you can explain that the dashboard is not the intelligence—it is the orchestrator that connects feature extraction, preprocessing, and machine learning into a single, user-friendly system.

Author's Note The next continuation will explain the dashboard itself, screen by screen. We'll cover: Every button. Every input field. Every output panel. Waveform visualization. Feature table. Prediction display. Confidence indicator. Error messages. Loading animation. What each UI element does internally. This chapter will prepare you to give a live demonstration of your project confidently during your viva or placement interview.

PART 2 — Complete Dashboard Walkthrough

Chapter 19 — Dashboard Architecture Part 2 — Complete Dashboard Walkthrough (Every Button Explained) Before Starting

In Part 1

we studied the architecture. Today, we'll study the dashboard itself. Imagine you are standing in front of your professor. The dashboard is open. He points to every button and asks "What does this do?" After this chapter, you'll be able to answer every question.

The Dashboard

Conceptually, our dashboard looks like this.

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Analog Fault Diagnosis System

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Upload CSV File

[Browse]

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Extract Features

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Predict Fault

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Predicted Fault:

RC_Open

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Feature Table

Peak RMS Mean. DC Level

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Waveform Plot

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The actual design may look different, but the workflow is the same.

Section 1 — Project Title

The first thing the user sees is the project title. Example Analog Circuit Fault Diagnosis Using Machine Learning Question. Does the title perform any computation? No. It simply identifies the application.

Section 2 — Browse Button

Now the first interactive element. Browse Question. What happens when the user clicks it? The operating system opens the file picker. The user selects the waveform CSV. The dashboard stores its file path. Nothing more.

PROFESSOR ASKS

What happens when the Browse button is clicked?

EXCELLENT ANSWER

The Browse button opens a file selection dialog, allowing the user to choose the waveform dataset that will later be processed for feature extraction and classification.

Step 3 — File Validation

Immediately after selection, the dashboard checks the file. Question. What is checked? ✔ File exists ✔ Correct extension ✔ Correct format ✔ Not empty ✔ Required columns present Question. Why? Because bad input causes bad output.

Step 4 — Feature Extraction Button

Next comes one of the most important buttons.

Extract Features

Question. Does this button perform Machine Learning? No. It only calls our feature extraction module. Pipeline becomes CSV

Feature Extraction

17 Features

Nothing else.

PROFESSOR ASKS

What happens when the Extract Features button is clicked?

EXCELLENT ANSWER

The dashboard invokes the feature extraction module, which calculates the 17 numerical features from the uploaded waveform and returns the feature vector.

Step 5 — Feature Table

After feature extraction, the dashboard shows all 17 features. Example **Peak** — 5.18. **RMS** — 3.42. **Energy** — 472. **DC Level** — 4.99. Question. Why display the features? Because users can verify that feature extraction worked correctly.

Engineering Advantage

Suppose one feature looks wrong. Example **Peak** — -100000. Question. Can we immediately notice the problem? Yes. The feature table helps debugging.

Step 6 — Predict Button

Now the main button.

Predict Fault

Question. What happens internally? Many things. Pipeline becomes

17 Features

StandardScaler

Load Saved Model

Prediction Everything is automatic. Question. Does the dashboard train the model? No. Only prediction.

PROFESSOR ASKS

What happens when the Predict button is clicked?

EXCELLENT ANSWER

The extracted features are first scaled using the saved StandardScaler. The trained SVM model is then loaded, the scaled feature vector is classified, and the predicted fault class is returned.

Step 7 — Prediction Display

Now the dashboard shows the result. Example

Predicted Fault

RC_Open Question. Where did this prediction come from? Not the dashboard. It came from the trained SVM. The dashboard only displays it.

Step 8 — Confidence Score (Optional)

Some implementations also show **Confidence** — 97.3%. Question. What does this mean? It indicates how confident the classifier is about its prediction. Higher confidence usually means greater certainty, but it does not guarantee correctness.

PROFESSOR ASKS

What is the purpose of displaying the confidence score?

EXCELLENT ANSWER

The confidence score indicates how strongly the classifier favors the predicted class compared to the other possible classes.

Step 9 — Waveform Plot

The dashboard also displays the waveform. Question. Why? Because the user can visually confirm the uploaded signal. Question. Does the waveform affect the prediction after feature extraction? No. Prediction uses only the extracted 17 features. The waveform is shown for visual understanding.

Step 10 — Status Messages

Good software always communicates with the user. Examples Loading. Extracting Features. Prediction Complete. Question. Do these affect Machine Learning? No. They improve user experience.

Step 11 — Error Messages

Suppose the user uploads an invalid file. Instead of crashing, the dashboard shows Invalid CSV Format. or No File Selected. Question. Why? Because professional software should fail gracefully.

Complete Dashboard Workflow

Everything we studied can now be summarized. **Browse** — Validate.

Extract Features

Display Features

Scale

Load Model

Predict

Display Result

Show Waveform

This is exactly what happens during one prediction.

Live Demonstration

Suppose your professor asks for a live demo. This is the sequence you should explain. "First, the user uploads a waveform CSV. The application validates the input, extracts the 17 features, scales them using the same StandardScaler used during training, loads the pre-trained SVM model, predicts the fault class, and finally displays the prediction along with the extracted features and waveform." This explanation is clear, professional, and complete.

Why This Dashboard Is Good

Notice something. The user doesn't need to know anything about Machine Learning. Everything is automated. That is the purpose of good software.

Connection to Our Project

Our complete dashboard workflow is now **User** — Browse. Validation

Feature Extraction

Feature Display

Scaling — SVM Prediction.

Display Result

Waveform Visualization

Everything fits together.

COMMON MISTAKES

MISTAKE: [X] The Browse button loads the SVM. Wrong.

MISTAKE: [X] The Predict button trains the model. Wrong.

MISTAKE: [X] The waveform is directly classified. Wrong. The classifier uses the extracted 17 features.

MISTAKE: [X] The dashboard contains the machine learning algorithm. Wrong. It simply calls the trained model.

REVISION NOTES

Remember
the dashboard sequence.
Browse — Choose File.
Extract — 17 Features.
Scale — Normalize.
Predict — SVM.
Display — Fault + Waveform.

MEMORY TRICK

Remember this sentence. "Browse, Extract, Scale, Predict, Display." Five words. Entire dashboard.

KNOWLEDGE CHECK

Level 1

- What does the Browse button do?
- What happens after feature extraction?
- Why is the StandardScaler used?
- Does the Predict button train the SVM?
- Why is the waveform displayed?

Level 2

- Explain the complete dashboard workflow.
- Why is input validation necessary?
- Why are extracted features displayed?
- How does the dashboard improve usability?
- Why is the trained model loaded instead of retrained?

Professor Level

- Explain every step performed after the user uploads a waveform.
- Why is separating the dashboard from the machine learning logic considered good software engineering?
- Why is the same StandardScaler used during deployment?
- How would you explain the dashboard during a live project demonstration?
- How does the dashboard convert a raw waveform into a fault diagnosis?
- End of Chapter 19 (Part 2)
- Congratulations.
- You now understand every important component of your dashboard.
- You can confidently explain each button, each processing step, and how the dashboard orchestrates the entire fault diagnosis pipeline from file upload to prediction.

Author's Note The next continuation will begin Chapter 20 — Limitations, Future Scope, and Possible Improvements. This chapter is extremely important because professors almost always ask: "What are the limitations of your project?" "If you had six more months, what would you improve?" "How can this be deployed in industry?" "Why didn't you use IoT, FPGA, or Deep Learning?" "How would you scale this system for real-time monitoring?" Answering these questions well often distinguishes an excellent project presentation from an average one.